Downscaling Urban Built-Up Surface from 1 km to 10 m Using EO Embeddings and Binary Settlement Priors

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Fine-resolution built-up maps are essential for urban analysis, exposure assessment, and spatial planning, yet many widely used built-up products remain available only at coarse resolution. Building on work in remote-sensing downscaling, dasymetric redistribution, and super-resolution mapping, we ask two questions: **(1) do Earth observation (EO) embeddings contain sufficient sub-kilometer information to reconstruct fine-scale urban built-up structure, and (2) does adding a binary settlement prior improve that reconstruction?** We address these questions in a prototype experiment for Accra, Ghana, by downscaling 2019 built-up surface from 1 km to 10 m (Atkinson, 2013).

Our framework combines a coarse mass-preserving constraint with fine-scale EO information. We use annual 10 m Google Satellite Embedding V1 features (Brown et al., 2025), reduced from 64 to 8 dimensions using PCA, together with the 10 m World Settlement Footprint (WSF) 2019 binary settlement (Marconcini et al., 2019) mask and GHSL built-up surface (Schiavina et al., 2023) products as coarse and fine reference layers. The learned models are lightweight fully convolutional networks with positive outputs. Training is driven by a coarse aggregation loss that matches the sum of 10 m predictions to the GHSL total within each 1 km cell, augmented by total-variation regularization and, in the guided model, a penalty on allocations outside WSF support. Final predictions are renormalized within each coarse cell to preserve the adjusted GHSL built-up total exactly.

We compare three increasingly informative strategies: a support-only redistribution over settlement pixels, an embeddings-only convolutional downscaler, and a guided model that combines embeddings with WSF-derived predictors and the outside-WSF penalty. The comparison reveals a clear division of strengths. The support-only approach sharply reproduces settlement morphology but acts largely as a literal redistribution of the binary mask. The embeddings-only model recovers broader intra-urban gradients and coherent large-scale spatial structure, demonstrating that EO embeddings alone contain meaningful fine-scale urban signal. The guided model combines these advantages: before renormalization it achieves a coarse-cell mean absolute error of about **5.4%** and RMSE of about **6.8%** relative to the mean adjusted 1 km built-up total, and after renormalization

allocates **93.5%** of predicted built-up mass within WSF support. Comparison against a 10 m GHSL reference shows that the guided model recovers substantially sharper urban form than the embeddings-only baseline while preserving city-scale intensity patterns better than the support-only benchmark.

These results show that EO embeddings and binary settlement priors are complementary for fine-scale urban downscaling and provide a practical route toward higher-resolution built-up mapping where continuous fine-scale labels are limited.

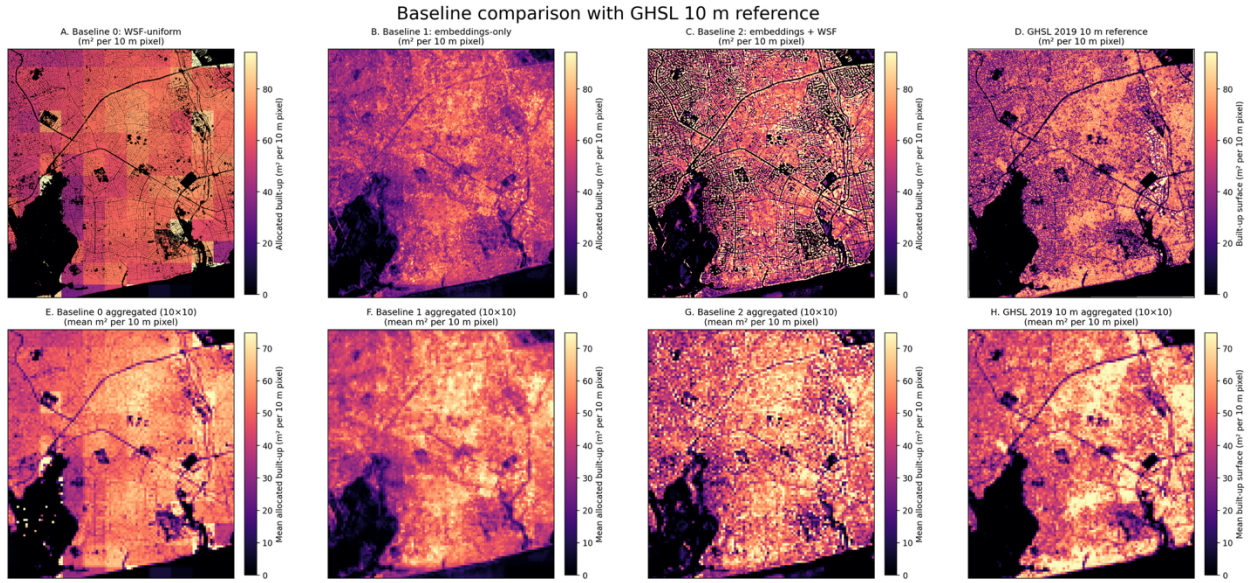

Figure 1: Comparison of three urban built-up downscaling baselines against a 10 m GHSL reference in Accra.

References

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